Create an ArrayList of String with the following members and delete an element with the index given as input

Add the following elements: [Volvo, BMW, Ford, Mazda]

Write a program to delete an element from the arraylist

Print the ArrayList

For example:

Input	Result
0	[BMW, Ford, Mazda]

Answer: (penalty regime: 0 %)

Language java

Reset answer

```java
import java.util.ArrayList;
import java.util.Scanner;
public class Main {
  public static void main(String[] args) {
      int n;
      Scanner sc = new Scanner(System.in);
      n = sc.nextInt(); //n is a value between 0 to 3
    ArrayList<String> cars = new ArrayList<String>();

    cars.add("Volvo");
    cars.add("BMW");
    cars.add("Ford");
    cars.add("Mazda");
    cars.remove(n);

    System.out.println(cars);
  }
}

```

Check

	Input	Expected	Got	
✓	0	[BMW, Ford, Mazda]	[BMW, Ford, Mazda]	✓

Passed all tests! ✓

Create an ArrayList aL1 of type String and perform the following operations.

1. Add “One”, “Two” into the arrayList aL1.
2. Read 5 elements from user.
3. Print the element at the index 5.
4. Change the element at index 2 as “Three”.
5. Create a printAll method which will print all the elements using an ListIterator.
6. Sort the elements and print the same.

For example:

Input	Result
Four Five Six Seven Eight	Element at index 5: Seven One Two Three Five Six Seven Eight [Eight, Five, One, Seven, Six, Three, Two]

Answer: (penalty regime: 0 %)

Language java

```java
import java.util.ArrayList;
import java.util.Collections;
import java.util.ListIterator;
import java.util.Scanner;

public class Demo{
    public static void printAll(ArrayList<String> list){
        ListIterator<String> li = list.listIterator();
        while(li.hasNext()){
            System.out.print(li.next()+" ");
        }
        System.out.println();
    }

    public static void main(String[]args){
        Scanner sc = new Scanner(System.in);

        ArrayList<String> AL1 = new ArrayList<>();
        AL1.add("One");
        AL1.add("Two");

        for(int i = 0;i<5;i++){
            AL1.add(sc.next());
        }
        System.out.println("Element at index 5: "+AL1.get(5));
        AL1.set(2,"Three");
        printAll(AL1);
        Collections.sort(AL1);

        System.out.println(AL1);
        sc.close();


    }
}
```

Check

	Input	Expected	Got
✓	Four Five Six Seven Eight	Element at index 5: Seven One Two Three Five Six Seven Eight [Eight, Five, One, Seven, Six, Three, Two]	Element at index 5: Se One Two Three Five Six [Eight, Five, One, Sev

Passed all tests! ✓

Krish has recently heard about something called Array Rotation and wants you to write a code for rotating an ArrayList. So what happens in arraylist rotation is, you are given an integer arraylist A of size N and an integer K, you have to rotate the arraylist in the right direction by K steps. The task is to print the rotated arraylist.

Input Format

The first line contains N, number of elements in the array and K number of steps.

The Second line contains N space-separated integers.

Output Format

For each test case on a new line, print the rotated array.

Sample Input:

5 2

1 2 3 4 5

Sample Output:

4 5 1 2 3

For example:

Input	Result
6 3 7 8 9 4 5 6	4 5 6 7 8 9
5 6 1 2 3 4 5	5 1 2 3 4

Answer: (penalty regime: 0 %)

Language java

```java

import java.util.*;

public class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int N = sc.nextInt();
        int K = sc.nextInt();
        ArrayList<Integer> A = new ArrayList<>();

        for(int i = 0;i<N;i++){
            A.add(sc.nextInt());

        }
        K = K % N;
        Collections.rotate(A,K);

        for(int i:A){
            System.out.print(i+" ");
        }


    }
}
```

Check

	Input	Expected	Got	
✓	5 2 1 2 3 4 5	4 5 1 2 3	4 5 1 2 3	✓
✓	5 6 1 2 3 4 5	5 1 2 3 4	5 1 2 3 4	✓

Passed all tests! ✓

Create an arraylist aL1 and perform the following operations.

1. Add elements 65, 32, 56 ,78, 96, 124
2. Remove the element at the index 4.
3. Add 45 at the index 2.
4. Check whether 78 is present in aL1. If yes, Print “True” Otherwise “False”
5. Print the ArrayList aL1.

Answer: (penalty regime: 0 %)

Language java

```
import java.util.*;

public class Demo{
    public static void main(String[] args){
        ArrayList<Integer> aL1 = new ArrayList<>();

        aL1.add(65);
        aL1.add(32);
        aL1.add(56);
        aL1.add(78);
        aL1.add(96);
        aL1.add(124);

        aL1.remove(4);

        aL1.add(2,45);

        if(aL1.contains(78)){
            System.out.println("True");
        }else{
            System.out.println("False");
        }
        System.out.println(aL1);
    }
}
```

Check

	Test	Expected	Got	
	/*TC 1*/	True [65, 32, 45, 56, 78, 124]	True [65, 32, 45, 56, 78, 124]	

Passed all tests!

Develop a java class Collections_ArrayList and perform the following operations.

1. Create a method saveEvenNumbers(int N) and use ArrayList to store even numbers from 2 to N, where N is a integer which is passed as a parameter to the method saveEvenNumbers(). The method should return the ArrayList (A1) created.
2. In the same class create a method printEvenNumbers() which iterates through the arrayList A1 in step 1, and It should multiply each number with 2 and display it in format 4,8,12....2*N.

Note: Create ArrayList A1 and A2 as instance variables.

For example:

Input	Result
23	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22] List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44]
78	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, List of Even Numbers multiplied by 2: [4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 88, 92, 96

Answer: (penalty regime: 0 %)

Language java

```java
import java.util.*;

public class Collections_ArrayList{

    ArrayList<Integer> A1 = new ArrayList<>();
    ArrayList<Integer> A2 = new ArrayList<>();


        public ArrayList<Integer> saveEvenNumbers(int N){
            A1.clear();
            for(int i = 2;i<=N;i+=2){
               A1.add(i);
            }
            return A1;
        }
        public void printEvenNumbers(){
            A2.clear();
            for(int num:A1){
                A2.add(num*2);

            }
            System.out.print(A2);
        }

public static void main(String[]args){
    Scanner sc = new Scanner(System.in);
    Collections_ArrayList obj = new Collections_ArrayList();
    int N = sc.nextInt();
    System.out.println("List of Even Numbers:");
    System.out.println(obj.saveEvenNumbers(N));
     System.out.println("List of Even Numbers multiplied by 2:");
     obj.printEvenNumbers();


}








}
```

Check

	Input	Expected
✓	23	List of Even Numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22] List of Even Numbers multiplied by 2:

Create an ArrayList aL1 of type Float and perform the following operations.

1. Read 5 elements from user.
2. Create an ArrayList aL2 and copy **all the** elements using one method from aL1 into aL2 and print the same using Iterator.
3. Remove **all the** elements from ArrayList aL1 using a single method.
4. Print the size of ArrayList aL1.

For example:

Input	Result
12.5 56.2 78.4 89.4 15.6	12.5 56.2 78.4 89.4 15.6 Size of aL1 is: 0

Answer: (penalty regime: 0 %)

Language java

```java
import java.util.*;

public class Demo{
    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);
        ArrayList<Float> aL1 = new ArrayList<>();

        for(int i =0;i<5;i++){
            aL1.add(sc.nextFloat());
        }

        ArrayList<Float> aL2 = new ArrayList<>();
        aL2.addAll(aL1);
        Iterator<Float> it = aL2.iterator();
        while(it.hasNext()){
            System.out.print(it.next()+" ");
        }
        System.out.println();

        aL1.clear();

        System.out.println("Size of aL1 is: "+aL1.size());
    }
}
```

Check

	Input	Expected	Got	
✓	12.5 56.2 78.4 89.4 15.6	12.5 56.2 78.4 89.4 15.6 Size of aL1 is: 0	12.5 56.2 78.4 89.4 15.6 Size of aL1 is: 0	✓

Passed all tests! ✓

Edit the ArrayList code given below so that the system prints the string in the appropriate index

For example:

Input	Result
0	Volvo

Answer: (penalty regime: 0 %)

Language java

Reset answer

```java
import java.util.ArrayList;
import java.util.Scanner;
public class Main {
  public static void main(String[] args) {
      int n;
      Scanner sc = new Scanner(System.in);
      n = sc.nextInt(); //n is a value between 0 to 3
    ArrayList<String> cars = new ArrayList<String>();

    cars.add("Volvo");
    cars.add("BMW");
    cars.add("Ford");
    cars.add("Mazda");
    String val;
    //type your line of code here..

    System.out.println(cars.get(n));
  }
}
```

Check

	Input	Expected	Got	
✓	0	Volvo	Volvo	✓

Passed all tests! ✓